RF Transmission, Reflections, and Standing Waves

Name: Durakhshan Ashraf Qadri

Roll Number: 261310606

1. Aim

The aim of this experiment is to study signal propagation, reflection and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

2. Objectives

1. To determine the phase velocity and velocity factor of the coaxial cable.
2. To determine the reflection coefficient and characteristic impedance.
3. To determine the inductance and capacitance per unit length.
4. To measure the insertion loss and attenuation of the cable.

3. Apparatus

RF signal generator, digital storage oscilloscope (DSO), coaxial cable, RF Tee connector, 50 Ω load, short-circuit termination and spectrum analyser.

4. Theory

A coaxial cable is a transmission line used to guide electromagnetic signals from one point to another. At radio frequencies, the cable supports transverse electromagnetic (TEM) propagation. The velocity of the signal in the cable is lower than the speed of light in vacuum because of the dielectric material between the conductors.

4.1. Phase Velocity

For a cable of length ℓ, the reflected signal travels a round-trip distance 2ℓ. Therefore, the round-trip time is

$$\tau = \frac{2\ell}{v_p}, \tag{1}$$

where v_p is the phase velocity.

Two consecutive maxima or two consecutive minima occur when the round-trip phase changes by 2π. Hence,

$$\Delta f = \frac{1}{\tau}, \tag{2}$$

and the phase velocity is

$$v_p = 2\ell\Delta f. \tag{3}$$

The velocity factor of the cable is defined as

$$VF = \frac{v_p}{c}, \tag{4}$$

where c is the speed of light in vacuum.

4.2. Reflection and Standing Waves

When a travelling wave reaches a load whose impedance Z_L differs from the characteristic impedance Z_0 of the cable, a part of the wave is reflected back towards the source. The incident and reflected waves superpose at the observation point. Depending on their relative phase, constructive and destructive interference produces maxima and minima in the measured signal amplitude. As the frequency is varied, the phase of the reflected wave changes, causing the measured amplitude to vary periodically.

For a load impedance Z_L, the voltage reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{5}$$

For an open circuit, $\Gamma = +1$, while for a matched load ($Z_L = Z_0$), $\Gamma = 0$. Thus, an open-ended cable produces strong reflections, whereas a $50\,\Omega$ termination should produce only a small reflection if the cable has a characteristic impedance close to $50\,\Omega$.

4.3. Reflection Coefficient from Measured Amplitudes

If $V_{\max}$ and $V_{\min}$ are the maximum and minimum amplitudes observed during a frequency sweep, the magnitude of the reflection coefficient can be estimated from

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{6}$$

This expression is used for both the open-circuit and $50\,\Omega$ termination measurements. A substantial reduction in $|\Gamma|$ for the $50\,\Omega$ termination indicates that the cable is approximately matched to $50\,\Omega$.

4.4. Inductance and Capacitance per Unit Length

For a low-loss transmission line, the phase velocity and characteristic impedance are related to the inductance L' and capacitance C' per unit length by

$$v_p = \frac{1}{\sqrt{L'C'}}, \qquad Z_0 = \sqrt{\frac{L'}{C'}}. \tag{7}$$

Therefore,

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}. \tag{8}$$

Using the measured phase velocity and the approximate value $Z_0 \approx 50\,\Omega$, the inductance and capacitance per unit length of the cable can be determined.

4.5. Cable Attenuation

As an RF signal propagates through a real cable, its power decreases because of losses in the cable. The insertion loss is determined by comparing the power measured using a short reference connection with the power measured after inserting the cable under test:

$$IL = P_{\text{ref}} - P_{\text{cable}}. \tag{9}$$

The attenuation per unit length is then

$$\alpha = \frac{IL}{\ell} \quad \text{dB/m}. \tag{10}$$

Thus, the experiment allows the phase velocity, velocity factor, reflection coefficient, characteristic impedance, inductance and capacitance per unit length, and attenuation of the coaxial cable to be determined experimentally.

5. Experimental Setup

The RF generator and oscilloscope were connected through an RF Tee connector, with the coaxial cable under test connected to the third port. The far end was successively left open, terminated with a 50 Ω load, and short-circuited for the reflection measurements. For the loss measurement, the cable was connected between the RF source and spectrum analyser.

6. Observations and Calculations

6.1. Part A: Phase Velocity and Velocity Factor

Measured cable length:

$$l = 0.91 \text{ m}.$$

Frequency spacing used:

$$\Delta f = 110 \text{ MHz}.$$

Therefore,

$$v_p = 2(0.91)(110 \times 10^6) = 2.002 \times 10^8 \ \mathrm{m\,s^{-1}}.$$

Hence,

$$VF = \frac{2.002 \times 10^8}{3 \times 10^8} \simeq 0.667.$$

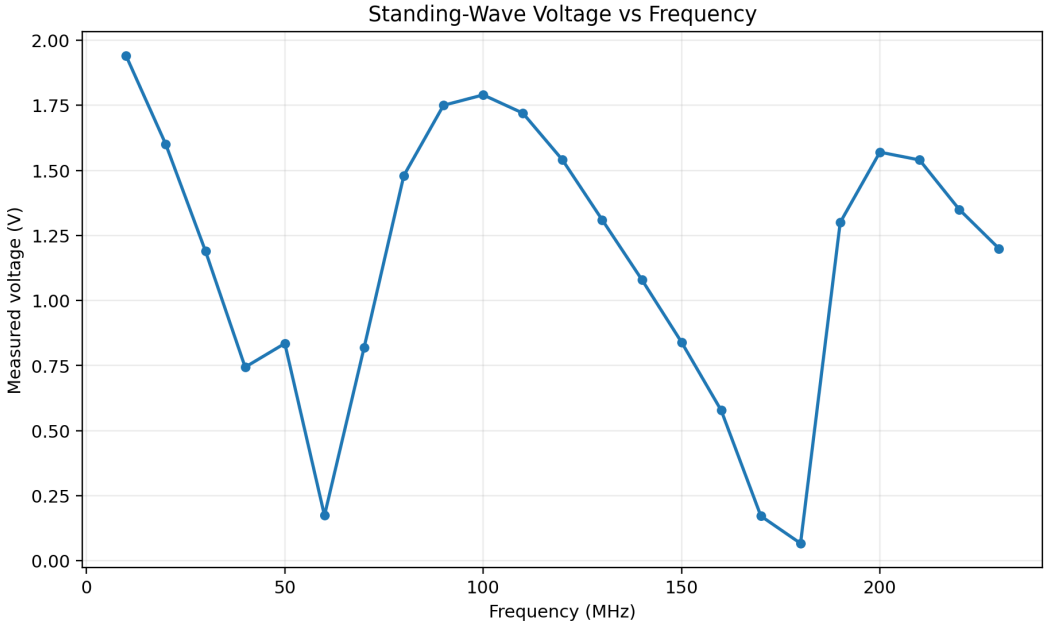


Figure 1: Measured standing-wave voltage as a function of frequency.

Table 1: Standing-wave voltage observations.

f (MHz)	V (V)	f (MHz)	V (V)
10	1.940	120	1.540
20	1.600	130	1.310
30	1.190	140	1.080
40	0.744	150	0.840
50	0.835	160	0.578
60	0.174	170	0.172
70	0.820	180	0.067
80	1.480	190	1.300
90	1.750	200	1.570
100	1.790	210	1.540
110	1.720	220	1.350
		230	1.200

6.2. Part B: Reflection Coefficient

The measured maximum and minimum voltages were:

Termination	$V_{\max}$	$V_{\min}$	$\lvert\Gamma\rvert$
Open circuit	920 mV	34 mV	0.929
50 Ω load	690 mV	626 mV	0.010
Short circuit	988 mV	76 mV	0.857

Table 2: Reflection-coefficient measurements.

For the open circuit,

$$|\Gamma_{\text{open}}| = \frac{920 - 34}{920 + 34} \simeq 0.929.$$

For the 50 Ω termination,

$$|\Gamma_{50}| = \frac{690 - 626}{690 + 626} \simeq 0.010.$$

For the short circuit,

$$|\Gamma_{\text{short}}| = \frac{988 - 76}{988 + 76} \simeq 0.857.$$

The very small reflection coefficient for the 50 Ω termination indicates that the load is closely matched to the transmission line.

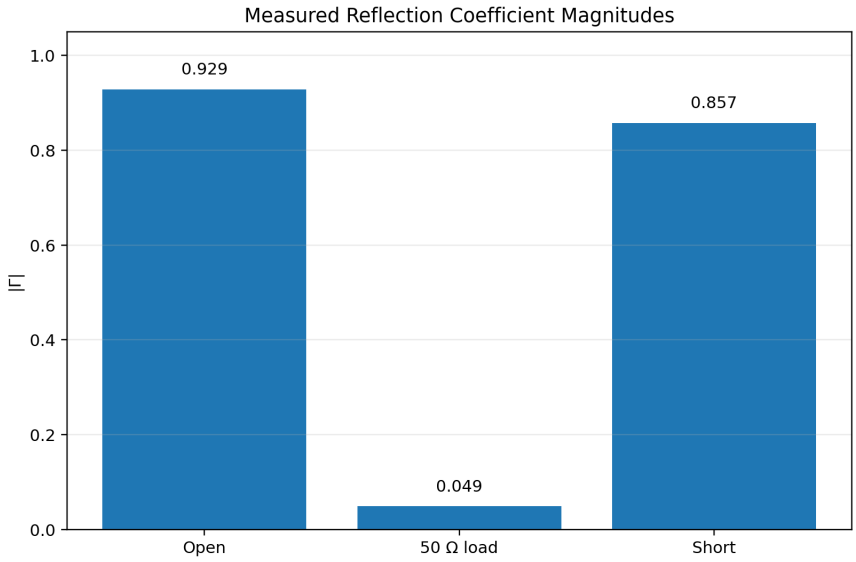


Figure 2: Comparison of measured reflection-coefficient magnitudes.

6.3. Part C: Inductance and Capacitance per Unit Length

Taking the nominal characteristic impedance as $Z_0 = 50\ \Omega$ and using $v_p = 2.002 \times 10^8\ \mathrm{m\,s^{-1}}$,

$$L' = \frac{50}{2.002 \times 10^8} = 2.498 \times 10^{-7}\ \mathrm{H\,m^{-1}} \simeq 249.8\ \mathrm{nH\,m^{-1}}.$$

Similarly,

$$C' = \frac{1}{50(2.002 \times 10^8)} = 9.99 \times 10^{-11}\ \mathrm{F\,m^{-1}} \simeq 99.9\ \mathrm{pF\,m^{-1}}.$$

$C' \simeq 99.9\ \mathrm{pF\,m^{-1}}$ $(0.10\ \mathrm{nF\,m^{-1}})$.

6.4. Part D: Cable Loss

The measured cable power levels were recorded in dBm. The graph below shows the measured power as a function of frequency.

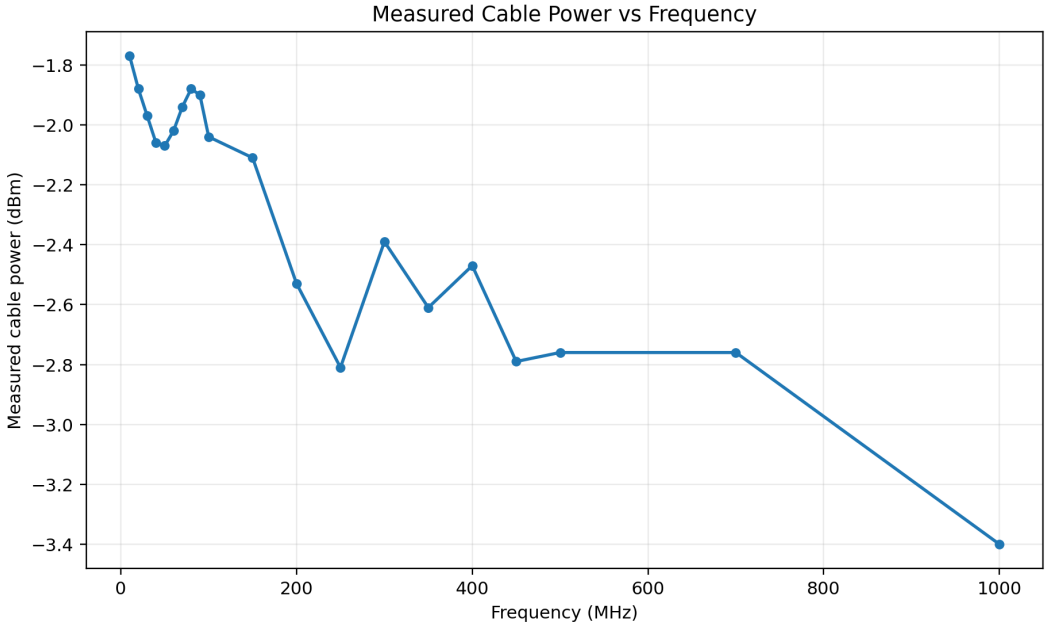


Figure 3: Measured cable power level as a function of frequency.

Table 3: Measured cable power.

f (MHz)	P (dBm)	f (MHz)	P (dBm)
10	-1.77	150	-2.11
20	-1.88	200	-2.53
30	-1.97	250	-2.81
40	-2.06	300	-2.39
50	-2.07	350	-2.61
60	-2.02	400	-2.47
70	-1.94	450	-2.79
80	-1.88	500	-2.76
90	-1.90	700	-2.76
100	-2.04	1000	-3.40

7. Results

Parameter	Experimental value
Cable length	0.91 m
Frequency spacing, Δf	110 MHz
Phase velocity, v_p	2.002×10^8 m/s
Velocity factor, VF	0.667
Open-circuit $\lvert\Gamma\rvert$	0.929
50 Ω load $\lvert\Gamma\rvert$	0.010
Short-circuit $\lvert\Gamma\rvert$	0.857
Inductance per unit length, L'	249.8 nH/m
Capacitance per unit length, C'	99.9 pF/m

Table 4: Summary of experimental results.

8. Conclusion

The experiment demonstrated reflection and standing-wave formation in a coaxial transmission line. The measured frequency spacing gave a phase velocity of approximately 2.00×10^8 m s^{-1} and a velocity factor of about 0.667. Strong reflection was observed for the open and short terminations, while the 50 Ω load produced a very small reflection coefficient of about 0.010, indicating good impedance matching. The distributed parameters were found to be approximately 249.8 nH m^{-1} and 99.9 pF m^{-1} for inductance and capacitance, respectively.

Statement on the Use of AI

A limited amount of AI was used in organizing the report, improving the clarity of explanations and formatting the document in LaTeX. However, the experimental measurements, observations, and numerical values reported in this work were obtained from my actual measurements performed during the experiment.